

MVP v1.0 Implementation Plan — Family Case Review

Status: Approved for execution · **Owners:** Engineering + Product · **Last updated:** 2026-08-29 **Companions:**

`../architecture/mvp_v1_system_design.md` (the architecture this plan builds — §ref'd throughout) ·

`../specifications/mvp_v1_prd.md` (US-x/FR-x/NFR-x/launch gates) ·

`../specifications/mvp_workflow_design.md` (S0–S7) · `../specifications/mvp_v1_engineering_notes.md` (ENG-1..11) · `../specifications/internal_operations_spec.md` (OPS/SRE) ·

`../specifications/product_roadmap.md` (v1.1–v4 — the post-MVP lens for every deprecation call)

1. Purpose and method

This plan turns the approved system design into a sequenced, estimated build with explicit acceptance criteria, and it settles the fate of every piece of legacy code in the repo. Two rules govern it:

1. **The system design's dependency order is the schedule's skeleton.** Nothing public ships before the security foundation (M0); the case-event spine (M1) precedes every feature that reads or writes case state.
2. **Deprecation decisions are made against the roadmap, not just against v1.0.** Code is deleted only when v1.1–v4 would rebuild it anyway (or when git history is a sufficient archive); code with a live future (ModelRouter/Ollama, MCP stdio servers, side-by-side viewer components, pgvector schema) is kept, parked, or config-gated — never casually removed. Section 5 is the full register.

2. Delivery approach and assumptions

- **Team:** 2 senior full-stack engineers + 1 product owner/founder; retained TX post-conviction counsel for the launch-gate review (~\$2–5k budgeted); 1 trained QA reviewer onboarded during M5.
- **Cadence:** two-week sprints; trunk-based development on `main` with short-lived feature branches; CI (already landed) grows the gates in §7 per milestone.
- **Environments:** local Docker Compose (Postgres+pgvector, Redis; Neo4j moves to an optional profile — §5); a single staging environment with Stripe test mode and a seeded reference-case fixture; production stood up in M6.
- **Definition of done (every story):** code + tests (integration tests hit real Postgres/Redis — the current mock-everything pattern is retired), audit events written where the spec requires, analytics events per the `snl.*` taxonomy, doc-diff landed in the owning spec, a11y/reading-level lint green on Daybreak surfaces.
- **Estimate discipline:** figures below are engineering-weeks (ew) of focused work, ranges honest. Total ≈ 36–42 ew → **14–16 calendar weeks with two engineers and the parallelization noted in §4** → launch-ready around **mid-December 2026**, with the attorney review and eval gate as the long poles to start early.

3. Milestones

Each milestone lists scope → key tasks → acceptance criteria (AC). System-design section references in parentheses.

M0 — Security foundation & repo hygiene (3 ew) — *blocks everything public*

Scope: eliminate the P0 defects (design §1) and clear the legacy underbrush (§5 register, “delete now” column).

- API JWT auth preHandler; remove all `x-tenant-id / x-user-id` header trust (§10.1). Typed web API client on `NEXT_PUBLIC_API_URL` (kills 14 hardcoded localhost sites).
- RLS: add `WITH CHECK` to every policy; route all tenant queries through `withTenant`; replace the mocked `rls.test.ts` with integration tests against real Postgres; fix the empty `20260607` migration.
- CORS pinned; Redis-backed rate limiting on auth/eligibility/presign/webhooks; Bull Board behind ADMIN.
- Audit service rewritten against real exports and wired into permissions/case mutations (first real `AuditLog` rows).
- Env cleanup (`GEMINI_API_KEY` unification); execute the “delete now” register entries; remove `socket.io` deps.
- ENG-10: move remaining `Test Case Files/` content to the encrypted eval bucket, land the manifest, decide the history rewrite (recommended: `git filter-repo` now, while clone count is low).

AC: an unauthenticated request to any tenant route is a 401; a forged-tenant JWT cannot read or **insert** cross-tenant rows (proven by integration test); repo contains no mocked route reachable in production; CI green.

M1 — Case spine (3 ew)

- `packages/case-lifecycle`: state enum, transition map, customer-visible mapping, event-type constants (§4).
- Schema migration set 1: `CaseStatus` /lane/vehicle/holds on Case, `CaseEvent` (append-only trigger), `EligibilityDraft`, checklist + upload-session tables (§5).
- Event append + projection helper (status column updated transactionally); SSE projector publishing the customer mapping; SSE route auth + case-access check (§9).

AC: illegal transitions throw; tracker stage for any case is derivable from events alone; SSE stream requires auth and never crosses cases; `docs.stalled_7d` derives from the stream.

M2 — Commerce & identity (4 ew)

- `CLIENT` role migration + middleware role matrix + API enforcement (§10.2; auth design §12).
- `/buy` flow API: disclosure-ack capture (archived per case — E-6), account+tenant creation, Stripe Checkout session; `payment.succeeded` webhook with `PaymentEvent` idempotency ledger; hourly reconciliation job (§7).

- `packages/email` (Resend + React Email): receipts, stage transitions, bounce/complaint webhooks → Ops queue. Password reset flow shipped (pulled into v1.0 per ENG-7).
- Eligibility draft endpoints (anon token, 30-day TTL, promote-and-delete).

AC: a purchase in Stripe test mode yields exactly one Case with disclosures archived, receipt sent, and events written — including when the webhook is dropped (reconciliation heals it); a CLIENT token cannot reach any staff surface; reset flow works end-to-end.

M3 — Intake pipeline (5–6 ew)

- Presign hardening (allowlist, size caps), multipart/resume sessions, HEIC conversion, ClamAV scan worker with quarantine messaging (ENG-4).
- Page normalization + perceptual/content-hash dedup; `DocumentPage` as the billable authority; meter endpoint with `duplicatesIgnored` (ENG-3); in-flow overage Checkout.
- OCR provider bake-off (Textract vs Document AI on the corpus's worst scans — design §12.3) then integration; per-page confidence; E-1 halt with customer options (re-upload / partial / refund).
- Interview → checklist generation; Tier-1 classification → echo-back events; `docs.complete` written once, SLA clock stamped via the business-day calendar service (ENG-9).

AC: a duplicate/shoebox re-upload never increments the meter; an infected file quarantines with plain-language messaging and never reaches ingestion; a low-OCR case halts **before** Tier-2 spend; records-complete fires exactly once per run.

M4 — Analysis pipeline (7–8 ew) — *start immediately after M1; runs parallel to M2/M3 against fixtures*

- Chunk-freeze per `AnalysisRun` (immutable, hashed, page-provenanced) (§6.1).
- `ModelRouter` tiers wired to real providers; batch submit + delayed-job polling + stage-budget watchdog with standard-API fallback (`BatchJob` , ENG-6); prompt-cache prefix stability verified via `usage.cache_read_input_tokens` .
- Screen set per lane (FR-1..5, FR-5a, FR-11); subsequent-writ §4-exception tagging; deadline engine on the shared calendar (FR-5 rules are launch-gate-reviewed by counsel).
- Tier-3 synthesis + Gemini adjudication; `Finding` / `FindingCitation` writes per ENG-2; grounding hard filter with excerpt-hash re-verification (FR-6/7).
- LangGraph rebuild: supervisor graph, real `ToolNode` loop, checkpointer registered; invoked only from the analysis worker.
- Cost telemetry: `CostRecord` on every model/OCR call incl. batch metadata; COGS query + \$54 alert (NFR-4).
- Eval harness integration: replayable per-screen outputs; both reference cases runnable from the bucket manifest.

AC: full pipeline run on the Gary reference case completes within stage budgets on batch pricing; every rendered finding's citations re-verify; adjudication disagreements carry the QA flag; per-case COGS is a single

query returning a number within budget.

M5 — QA console & report (5 ew)

- `/qa` queue (adjudication-flagged first), side-by-side verify with click-through to frozen page images, edit (provenance → `ai_human_edited`, audit-logged), approve/reject; approve writes the findings snapshot + template version (ENG-8/ENG-11).
- `packages/reports`: Part A/B React templates (i18n-keyed, `es` stubbed), Chromium tagged-PDF render, reading-level lint on Part A strings; deadline/tolling copy exactly per PRD FR-5 language rules.
- Cited-page image pre-render at approval; share-with-a-lawyer links (hashed token, expiry, revocation, access log); delivery flow with `DELIVERED` -on-confirmation and bounce → Ops (§8, §9).
- Re-run diff on `Finding.stableKey` (US-6).

AC: a QA reviewer completes a reference case in ~30 min; nothing customer-visible exists pre-approval; a template change after approval does not alter an in-flight report; a bounced delivery email leaves the case `READY` and opens an Ops item.

M6 — Ops console, observability, analytics (5 ew)

- `/ops` per OPS-1..7: case queue with stall/deadline flags, audited Stripe-linked refunds computed from the page authority, disclosure-archive export, retention/deletion workflow (hard-delete incl. S3 + entity rows + report artifacts), delay-ours with honest tracker copy, reviewer management on existing RBAC.
- Sentry + OTel across web/API/workers; pipeline-health board; SRE-2 alert routing; SRE-4 runbooks written (stall, provider outage/batch fallback, S3, webhook loss, low-OCR cohort).
- Self-hosted PostHog + `snl.*` events wired everywhere; server-side mirrors for money/pipeline facts; `packages/config` flags with audited admin UI (PMX-2); weekly funnel report automated (PMX-1).

AC: every promise in the customer specs (refund, deletion, delay honesty, SLA authority) is executable from `/ops` with an audit row — zero manual SQL paths remain; a simulated provider outage produces the truthful tracker state, never fake progress.

M7 — Launch hardening & gates (3–4 ew, overlaps M5/M6)

- End-to-end dry runs on both reference cases: purchase → upload → report → refund path (gate 4).
- Eval green per NFR-1 on both cases (gate 1); attorney review package delivered (templates, copy, T&C, US-0 routing, deadline rules, §4 handling, R-6 structure — gate 2); E&O bound (gate 3).
- Load check at NFR-5 (10 cases/day); a11y (WCAG 2.1 AA) + reading-level CI on all Daybreak routes; pentest of the tenant boundary (SRE-6).
- Open product decisions closed (owner: PO): SLA `N`, plea-lane pricing, payment plans, R-6 referral structure, brand/R-5.

AC: all five PRD §7 launch gates checked; go/no-go review documented.

Daybreak UX build (runs M2 → M5 as a parallel track)

The UI is spec-complete (`mvp_ui_design_spec.md` , `landing_page_spec.md`); this plan deliberately sequences its build **after the contracts it consumes exist**: landing + S0 wizard on M1’s draft API; buy flow on M2; checklist/upload/echo-back/meter on M3; tracker on M1’s SSE mapping; interstitial/report/next-steps on M5’s report data. Token package (`db-*` / `hg-*`), externalized copy canon, and a11y CI (UXG-1..5) land with the first Daybreak screen. UX estimate: ~6 ew inside the M2–M5 window (it is the second engineer’s parallel lane at several points).

4. Sequencing at a glance

Sprint (2 wks)	Track A	Track B
S1	M0 security foundation	M0 hygiene/deletions + eval-bucket move
S2	M1 case spine	M4 start: chunk-freeze + router on fixtures
S3	M2 commerce/identity	M4: screens + batch orchestration
S4	M3 intake (upload/scan/dedup)	M4: synthesis + adjudication + grounding
S5	M3 intake (OCR/checklist/clock) + Daybreak S0–S3 UI	M4 finish + eval harness runs
S6	M5 QA console + report	M6 ops console start; Daybreak tracker/report UI
S7	M6 observability/analytics	M7 dry runs, eval, counsel package
S8	M7 gates, pentest, buffer	Launch go/no-go

Counsel review (gate 2) is commissioned at S5 so findings land before S8. The eval ledger (attorney-signed ground truth) is commissioned at S1 — it gates M4’s exit and takes calendar time we don’t control.

5. Legacy code disposition register (the deprecation plan)

Decisions reflect the post-MVP roadmap: **v1.1** Spanish/AV/records-concierge/attorney-signed add-on · **v2** clinic tier (multi-seat, triage dashboard) · **v3** professional workspace (chat, graph UI, drafting, DOCX export, Clio/PACER, subscriptions) · **v4** Sovereign self-host (Ollama-primary). “Delete” means removed from the tree in M0 — git history is the archive; nothing of production value is lost.

5.1 Delete now (mocked, broken, or duplicative — v-next would rebuild anyway)

Asset	State	Rationale / post-MVP reflection
<code>apps/api/src/routes/chat.ts</code>	Unregistered; mocked; targets wrong Fastify/WS API	v3 chat is SSE/LangGraph-based per design — zero salvage value
<code>apps/api/src/routes/export.ts</code> + duplicate <code>export.ts</code> / <code>export.service.ts</code>	Hardcoded mock claims → real docx	v1.0 reports are PDF; v3 DOCX master-sheets will be regenerated from the Finding schema, not this route. Drop <code>docx</code> dep until v3
<code>apps/api/src/routes/webhooks.ts</code> (PACER)	Zod + <code>console.log</code>	v3 integration; stub misleads
<code>apps/api/src/services/document-processor.worker.ts</code>	Orphan; doesn't compile	Superseded by M3/M4 workers
<code>apps/api/src/services/storage.service.ts</code>	Unused duplicate of upload route	Consolidated into one storage module in M3
<code>apps/api/src/workers/media.worker.ts</code> + <code>media queue</code>	Never enqueued; latent 1536 → 768-dim crash	v1.1 A/V add-on rebuilds on the Phase-5 Whisper/FFmpeg design with metering — this mock isn't the seed of it
Mocked chunking/extraction in <code>ingestion.worker.ts</code> / <code>entity.worker.ts</code>	Hardcoded 2-chunk array; <code>.includes()</code> “extraction”	Replaced wholesale by M3 intake + M4 analysis workers
<code>apps/web workspace chat</code> mock + <code>/workspace/[caseId]/graph</code> page + hardcoded KnowledgeGraph data	<code>setTimeout</code> fake; fixture nodes	v3 surfaces; fake data on real routes is a trust liability. Component exceptions kept — see 5.2
<code>/api/auth/cliio/callback</code>	Returns a fake token, redirects to a nonexistent route	v3 <code>mcp-cliio-sync</code> does OAuth properly
<code>packages/ai</code> LangChain tool wrappers (<code>mcp-tx-case-law</code> , <code>bluebook</code> , <code>cliio</code> , <code>axon</code> , <code>statutes-pro</code>)	Canned JSON (“Ex parte Smith”)	The real stdio MCP servers are kept (5.2); canned wrappers actively poison eval work
<code>socket.io</code> + <code>fastify-socket.io</code> deps	Declared, never imported	Real-time is SSE-only (design §12.5)

5.2 Keep — parked, config-gated, or repurposed (a live post-MVP future)

Asset	v1.0 role	Post-MVP future
<code>packages/auth</code> (all of it)	Production auth, extended per §12 addendum	Every tier
<code>ModelRouter</code> (<code>packages/ai/router.ts</code>)	Active, Ollama config-off	Ollama returns at ≥150 cases/mo (v1.1 ops) and is the v4 Sovereign engine
Real stdio MCP servers (<code>tx-statutes</code> , <code>tdcj-policy-expert</code> , <code>tx-appellate-rules</code> , <code>tx-jury-charges</code> , <code>dev-tools</code>)	Pipeline tools where the eval shows benefit	v3 drafting + v4 local MCP stack
<code>PostgresCheckpointSaver</code> + <code>GraphState</code> table	Fixed and registered in M4 (not deleted)	Long-running professional workflows v3
<code>ParchmentViewer</code> , <code>react-resizable-panels</code> side-by-side shell	QA console source pane (M5)	The v3 attorney workspace — v1.0 dogfoods it daily
<code>ViabilityScorecard</code> component	Dormant	v2 clinic triage dashboard; shares the <code>FindingCard</code> data contract
pgvector schema (<code>DocumentChunk.embedding</code>)	Embedding generation paused (written-never-read costs money); column kept	v3 semantic search/chat revives it with an HNSW index
Neo4j (<code>neo4j.ts</code> , docker service)	Out of the v1.0 runtime; compose moves it to an optional <code>graph</code> profile; entity index lives in Postgres (design §12.1)	v2/v3 graph surfaces re-adopt with tenant-scoped labels
Staff pages (<code>/dashboard</code> , <code>/dashboard/permissions</code> , <code>/dashboard/account</code> , <code>(auth)</code> group)	Live staff/QA administration	v2 multi-seat tenancy builds on them
<code>dev-cli.ts</code> , <code>index-codebase.ts</code> , LanceDB dev-RAG	Developer tooling	Unchanged
<code>mcp-bluebook-sanitizer</code> (regex placeholder)	Dormant	v3 drafting; not worth deleting, not worth improving yet

5.3 Docs deprecation

`website_architecture.md` already carries the supersession banner (platform/v3 reference).

`full_implementation_plan.md` + `execution_roadmap.md` + `ux_buildout_plan.md` remain the historical engineering-build record — **this document supersedes them for all forward work**; a status note is added to

each. `mcp_integration_strategy.md` 's “Ollama as orchestrator” framing is superseded by the routing revision (already noted there via `cost_optimization_ollama.md`).

6. Risks and mitigations

Risk	Likelihood/Impact	Mitigation
Eval ledger (attorney-signed ground truth) arrives late — gates M4 exit	Med / High	Commission at S1; treat as external dependency with weekly check-ins; harness runs against draft ledger meanwhile
Batch-API worst case (24h) breaks SLA arithmetic	Med / Med	ENG-6 watchdog + standard-API fallback built in M4, not bolted on; SLA N set after measuring reference-case runs
OCR quality on aged scans below viability	Med / High	S1 bake-off on the corpus's worst pages; E-1 halt protects refund economics either way
Prompt-cache misses silently triple Tier-2 cost	Med / Med	Cache-hit telemetry is an SRE-2 alert from day one; prefix byte-stability is a unit test
Counsel review forces copy/flow rework late	Med / Med	Deadline-engine rules, US-0 routing, and disclosure copy sent at S5, not S8
Two-engineer team: M4 complexity concentrates in one head	High / Med	Design doc §6 is the shared contract; pipeline PRs reviewed cross-track; eval harness is the objective referee
Chargeback wave from unmet expectations	Low / High	W-2 disclosure archive + OPS-3 export ship with commerce (M2), not later; 5% reserve budgeted
History still contains real case records	Certain / Med	M0 filter-repo decision — do it before more clones exist

7. Launch-gate traceability

PRD §7 gate	Delivered by
1. Eval harness green (both reference cases)	M4 + M7 (ledger from S1)
2. Attorney review (copy, T&C, routing, deadlines, §4, R-6)	Commissioned S5, closed M7
3. E&O insurance bound	PO track, M7
4. QA console + trained reviewer + E2E dry run	M5 + M7
4a. Ops console, Sentry/OTel, rate limiting live	M0 (limits) + M6
5. Billing spec implemented (Checkout, cap, overage, refunds)	M2 + M3 + M6

8. Governance

- **Eval-gated model changes:** no router/model/prompt change ships without the `model_evaluation.md` decision rules (per-category recall + citation fidelity, no regression).
- **Doc-diff discipline:** every shipped behavior change lands a diff in its owning spec in the same PR (analytics plan §6); this plan is updated at each milestone review.
- **Red lines:** disclosures, not-legal-advice language, interstitial honesty, and urgency mechanics are never A/B tested (analytics plan §4.3); all consumer-visible copy passes R-5 tone review; anything legal-adjacent passes counsel.
- **Scope guard:** the PRD §2 non-goals are enforced in review — no chat, no customer graph, no research claims, no subscriptions, no pro se pathways. Feature pressure routes to the roadmap, not the sprint.